



A model of macroalgal decomposition

Journal:	<i>Journal of the Royal Statistical Society: Series C</i>
Manuscript ID	Draft
Manuscript Type:	Original Article
Date Submitted by the Author:	n/a
Complete List of Authors:	Wright, Luka; UWA Oceans Institute, ; UWA School of Biological Sciences,
Keywords:	Bayesian statistics, decay, decomposition, degradation, exponential model, mathematical ecology

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A model of macroalgal decomposition

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Running head

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Abstract

Macroalgae are currently the focus of blue carbon, biological carbon dioxide (CO₂) removal in the ocean. Carbon stored in seaweed forest biomass is inconsequential to CO₂ removal for climate change mitigation. However, carbon exported as seaweed detritus represents ~7.5% of anthropogenic CO₂ emissions annually. Macroalgal blue carbon hinges on the fate of this detritus, so a reliable model of its decomposition is key. Macroalgal decomposition has been neither consistently nor reliably modelled. Here I present the first model tailored to macroalgal decomposition. Informed by the theory of detrital photosynthesis, it treats detritus as an entity in limbo that can initially either grow or decompose. This is achieved by replacing the rate constant of the classic exponential decay model with a logistic function of time, mirroring detrital photosynthesis. The resulting model has three parameters that describe the initial relative growth rate, the half-life of detrital photosynthesis, and the final exponential decay rate which is comparable to the rate constant of the classic model. Using a Bayesian implementation in R with Stan, I show that the model performs well given empirical data from a variety of contexts. I anticipate that it will prove instrumental to resolving the debate around macroalgal CO₂ removal.

Keywords

Bayesian statistics, decay, decomposition, degradation, exponential model, mathematical ecology

Introduction

In a seminal paper on mathematical ecology, Olson (1963) popularised the ordinary differential equation of leaf litter decomposition

$$\frac{dm}{dt} = -km \quad (1)$$

which describes the rate of change in litter mass m proportional to itself, a relationship which is parameterised by the decay constant k . This reflects our ecological understanding that given a theoretically limitless pool of decomposers, decomposition is proportional to the detrital pool. The equation integrates to

$$\ln \frac{m(t)}{m_0} = -kt \quad (2)$$

and simplifies to the familiar exponential decay function

$$m(t) = m_0 e^{-kt} \quad (3)$$

where $m(t)$ is a function of the initial litter mass m_0 , the decay constant k and time t . This function has prevailed as the model of choice for decomposition.

Interest in decomposition has been rekindled due to the hope that plants can mitigate anthropogenic climate change by fixing carbon dioxide (CO₂), assimilating it into carbohydrates, and sequestering it in their biomass. Carbon sequestration timescales of decades to centuries are being considered the minimum and ten millennia or longer are preferred (Babiker et al., 2022). Plant biomass can be split into two distinct carbon sinks: living plants and the detrital pool, of which the latter is considered the more promising. Often surprising to plant ecologists, plant detritus amounts to a biomass that is generally comparable and often far greater than that of living plants (Cebrián & Duarte, 1995, Wright & Kregting, 2023). But more importantly, living plants, even when considering stable or increasing populations, are limited by biomass-density constraints (Creed et al., 2019) and lifespans of typically less than a millennium (Thomas, 2013). Detritus on the other hand is believed to have the potential of cumulative sequestration over much longer timescales. Fossil fuels were created via the Carboniferous detrital pathway, mostly of scale trees

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3 67 (Hibbett et al., 2016), and the hope of biological CO₂ removal essentially boils down to re-
4 68 enacting fossil fuel generation.
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8 70 Biological CO₂ removal in the ocean, which amounts to somewhat less than on land
9 71 (Friedlingstein et al., 2025), is referred to by the shorthand blue carbon (Babiker et al., 2022).
10 72 Macroalgae have recently moved into the focus of blue carbon (Pessarrodona et al., 2023),
11 73 primarily due to their substantial global area (Duarte et al., 2022), biomass packing (Creed et
12 74 al., 2019), productivity (Pessarrodona et al., 2022) and detrital export (Pessarrodona et al.,
13 75 2023, Filbee-Dexter et al., 2024). Notably, decomposition is disproportionately consequential
14 76 for macroalgal blue carbon. Unlike their fellow blue carbon ecosystems—seagrass meadows,
15 77 mangrove forests and salt marshes—seaweed forests cannot accumulate carbon in below-
16 78 ground biomass and soil. There is thus a set limit for the living seaweed carbon sink which
17 79 can be estimated as the product of 6.1 million km² global cover (Duarte et al., 2022), 1.2 ± 3
18 80 kg m⁻² (mean $\pm$ standard deviation) standing biomass (Creed et al., 2019) and 28 ± 4.5 %
19 81 carbon content (Pessarrodona et al., 2022). This yields 0.34 ± 0.87 kg C m⁻² and 2.1 ± 5.3 Gt
20 82 C globally, or only 19 ± 49 % of annual anthropogenic CO₂ emissions (Friedlingstein et al.,
21 83 2025). However, seaweeds are thought to assimilate 0.32 ± 0.54 kg C m⁻² y⁻¹ (Pessarrodona
22 84 et al., 2022) and 1.3 Gt C y⁻¹ globally (Duarte et al., 2022) of which 61 ± 4.2 % is exported
23 85 as detritus (Pessarrodona et al., 2023). Hence 0.19 ± 0.33 kg C m⁻² y⁻¹ and $0.81 \pm 0.055^*$ Gt
24 86 C y⁻¹ globally, or $7.5 \pm 0.51^*$ % of annual anthropogenic CO₂ emissions, may enter the
25 87 detrital pool yearly. Viewed cumulatively this is clearly the more substantial carbon sink.
26 88 Decomposition alone therefore determines whether macroalgal blue carbon is significant or
27 89 not, as is illustrated by the central role it takes in the latest macroalgal blue carbon models
28 90 (Filbee-Dexter et al., 2024). Despite this, macroalgal decomposition remains comparatively
29 91 poorly understood.
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31 92
32
33 93 Phycologists have never been able to properly model macroalgal decomposition.
34 94 Decomposition trajectories rarely match Equation 3 reasonably well (e.g. de Bettignies et al.,
35 95 2020, Pedersen et al., 2021). Most studies have therefore either refrained from fitting a model
36 96 (e.g. Birch et al., 1983, Hamersley et al., 2015) or most commonly fit linear models (e.g.

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58 * This standard deviation is an underestimate because no uncertainty was provided for the
59 global primary production estimate (Duarte et al., 2022).
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Bedford & Moore, 1984, Brouwer, 1996, Frontier et al., 2021, Filbee-Dexter et al., 2022, Frontier et al., 2022, Wright et al., 2022). Meta-analyses of macroalgal k (Equation 3) are numerous (Enríquez et al., 1993, Banta et al., 2004, Cebrián & Lartigue, 2004, Filbee-Dexter et al., 2024, Kennedy & Blain, 2025, White & Norkko, 2025) but have generally assumed Equation 3 by default and not assessed the validity of this model in the macroalgal context. It is evident that these meta-analyses are often hampered by the complexity and diversity of macroalgal decomposition trajectories (Kennedy & Blain, 2025, White & Norkko, 2025). The only serious attempts at modelling diverse trajectories have invariably invoked the logistic function alongside the exponential (Bourguès et al., 1996, Kennedy & Blain, 2025). This defeats the purpose of a model since comparing logistic and exponential rate constants is as effective as squinting at the data. Kennedy & Blain (2025) realised this when trying to compare half-lives ($\frac{\ln 2}{k}$ for Equation 3) across exponential and logistic models and ultimately resorted to forcing an exponential model to derive this parameter, which undermines fitting the logistic model in the first place.

We now know what causes the observed diversity in macroalgal decomposition trajectories. Early phycologists noticed the need to “pre-kill” macroalgal detritus (Birch et al., 1983, Smith & Foreman, 1984, Rieper-Kirchner, 1989, Brouwer, 1996) if it was to behave according to the expectations of the classic model (Equation 3). Recently it became evident that this is because excised macroalgal tissue can continue to photosynthesise for months (de Bettignies et al., 2020), which may allow macroalgal detritus[†] to grow rather than decompose over the short term (Frontier et al., 2021, Pedersen et al., 2021, Filbee-Dexter et al., 2022, Frontier et al., 2022, Wright et al., 2022). There is now extensive investigation into what has since been termed detrital photosynthesis (Frontier et al., 2021, Wright & Foggo, 2021, Frontier et al., 2022, Wright et al., 2022, Wright & Kregting, 2023, Wright et al., 2024) and macroalgae have been found to be unique among plants in having detritus that photosynthesises to an extent that is likely to influence its decomposition (Wright, in review). This knowledge enables the creation of a model that accurately describes macroalgal

[†] I have faced semantic challenges when communicating this phenomenon since detritus has traditionally been used only to refer to dead matter. To avoid confusion, I have suggested “detritus” for all detritus and “detritus *sensu stricto*” for dead detritus (Wright, in review). I use this terminology hereinafter.

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3 125 decomposition by taking into account the effect of detrital photosynthesis. It is probable that,
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5 126 ignorant of detrital photosynthesis, previous models have substantially underestimated
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7 127 macroalgal decomposition with obvious consequences for blue carbon accounting. To
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9 128 remedy this I here present a model with only three parameters that is tailored to the unique
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11 129 case of macroalgal decomposition. First I describe its mathematical form, then I implement it
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13 130 as a Bayesian statistical model in R (R Core Team, 2022) with Stan (Carpenter et al., 2017),
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15 131 and finally I validate the model against a variety of empirical examples.
16

17 133 **The mathematical model**

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19 134 *...tossed and swept like dead leaves from one spot to another, never resting, never giving*
20
21 135 *back their goodness to the earth, never fully dead but in some vast limbo between life and*
22
23 136 *extinction, tossing and tumbling without end...* — Philip Pullman (1994) *The Tin Princess*
24

25
26 138 Macroalgal detritus is essentially in limbo. It may grow or decay, but not indefinitely. The
27
28 139 goal is to model this temporary limbo. There are several possible approaches to describing
29
30 140 this mathematically. As mentioned above, the logistic function has been the only attempt so
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32 141 far, but its rate constant is not comparable to that of Equation 3 and it cannot describe the full
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34 142 spectrum of possible decomposition trajectories (e.g. an increase followed by a decrease).
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36 143 Hence we must dismiss it. Another approach could be a composite function of a saturating
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38 144 function (e.g. rectangular hyperbola, hyperbolic tangent, exponential) and Equation 3. This
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40 145 may describe all possible trajectories but is somewhat of an artificial construct and more
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42 146 importantly would have more parameters than is practical. The best way forward is to stick
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44 147 with our trusty Equation 3 but rebuild it from its differential form (Equation 1). Instead of
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46 148 treating k as constant, we can let it be a function of time:
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51 150
$$\frac{dm}{dt} = k(t)m \tag{4}$$

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54 152 To my knowledge this approach has only been applied twice before, by Rovira & Rovira
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56 153 (2010) to describe the variety of terrestrial leaf litter decomposition trajectories and by
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58 154 Manojlovic (2021) to model seagrass decomposition (Trevathan-Tackett et al., 2020). Rovira
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60 155 & Rovira (2010) chose the form of $k(t)$ based on context while Manojlovic (2021) used a
156
157 156 random walk, the simplest autoregression which defines k only in terms of k at the previous
timepoint plus normal error:

$$k(t) \sim N(k(t - \Delta t), \sigma) \quad (5)$$

where σ is the standard deviation of the normal distribution around the mean $k(t - \Delta t)$ and determines the k step size of the random walk. The timestep size Δt is typically set to 1. While this approach is the most flexible, it requires sophisticated Bayesian state space modelling with an ordinary differential equation solver. Manojlovic (2021) used LibBi (Murray, 2015) and to my knowledge there is currently no way to implement this in the more popular Stan language (Manojlovic, personal communication, discourse.mc-stan.org/t/random-walk-in-rstan/34308). Furthermore, being the simplest type of order 1 autoregressive process, a random walk's only parameter is σ which has no inherent directionality and precludes informative prediction beyond the range of the data. Finally, no comparison can be made between studies beyond overlaying plots of $k(t)$, and formal comparison with studies that use the traditional model (Equation 3) is out of question. A model with informative parameters which can be formally compared is the way forward. Hence I here opt for the path laid out by Rovira & Rovira (2010).

Now the most important task is to find an appropriate function to describe k . Since the underlying process affecting k is detrital photosynthesis, it makes sense to start there. The change in detrital photosynthesis with time post excision, or detrital age, has mostly been described using various linear models (de Bettignies et al., 2020, Frontier et al., 2021, Wright & Foggo, 2021, Frontier et al., 2022, Wright et al., 2022, Wright & Kregting, 2023, Wright et al., 2024). In truth there is no reasonable mechanism for photosynthesis to increase post excision[‡], but it is often maintained near maximal levels until a breakpoint is reached and it decays to a minimum beyond which there logically is no further decline. Voilà, logistic decay! In Wright (in review) I have described detrital photosynthesis (p) as

$$p(t) = \frac{\alpha + \tau}{1 + e^{r(t - \mu)}} \tau \quad (6)$$

[‡] We sometimes see this due to some unrelated factor such as a change in light environment (Frontier et al., 2021, Wright & Kregting, 2023).

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187 where α and τ may be read as the maximal CO₂ assimilation and production of the plant and
188 its detritus respectively, r is the logistic rate of decay and μ is the time of photosynthetic
189 death, more specifically the half-life of photosynthesis. The naming of parameters is inspired
190 by the word מָת (graecised $\tau\mu\alpha$), meaning truth. Composed of the first, middle and last letters
191 of the Hebrew alphabet, it symbolises life ($\aleph$), transition (μ) and death (τ). In legend this word
192 lends life to the golem, but removing $\aleph$ changes its meaning to death (מָת) and the golem dies.
193 As in the story of the golem, the logistic function is optimal for modelling a shift between
194 two alternate states and is thus the logical choice here.

196 In practice, I found the full four-parameter form of Equation 6 difficult to work with because
197 r and μ are not identifiable and the model affords little control over the intercept, for which
198 we known $p(0) \approx \alpha$ to be true. This usually requires a workaround, such as parameterising the
199 model in terms of the logistic intercept ($r\mu$) or constraining the model with a joint prior on $\ln r$
200 + $\ln \mu$ (Wright, in review). There are two ways to simplify Equation 6 to a three-parameter
201 form: (1) set r constant and (2) describe r relative to μ , which is effectively the same as
202 setting $r\mu$ constant. Reasonable values are $r = 1$ for option 1, which is roughly equivalent to
203 photosynthetic death over one week, and $r = \frac{5}{\mu}$ for option 2 which implies that $r\mu = 5$, so $p(0)$
204 = $0.99(\alpha + \tau) - \tau \approx \alpha$. Theory would favour option 2 because we cannot assume a universal
205 logistic rate of decay for detrital photosynthesis but we know $p(0) \approx \alpha$. Nonetheless, option 1
206 allows rapid photosynthetic decay independent of μ while option 2 implies that if detrital
207 photosynthesis persists it must be due to slow decay initiated at $t = 0$. Accordingly the
208 interpretation of α changes from detrital photosynthesis until photosynthetic death in option 1
209 to initial photosynthesis in option 2. While we know that the ultimate cause of photosynthetic
210 decay is that initiating event (e.g. excision) and this agrees well with detrital photosynthesis
211 trajectories (Wright, in review), data on macroalgal decomposition often seem to suggest that
212 detrital photosynthesis is unperturbed until some more proximate mechanism causes it to fail
213 (de Bettignies et al., 2020, Frontier et al., 2021, Frontier et al., 2022). I therefore tested both
214 decomposition models on two empirical datasets (Brouwer, 1996, Frontier et al., 2022) given
215 identical priors to compare performance. Option 2 has marginally better predictive
216 performance according to leave-one-out cross-validation (Vehtari et al., 2017) but more
217 importantly has smoother posteriors, while option 1 shows signs of bimodality, non-
218 identifiability or jaggedness in posteriors (github.com/lukaseamus/limbodeco). Thus I
219 proceed with option 2 to describe k over time as

220

$$k(t) = \frac{\alpha + \tau}{1 + e^{\frac{5(t-\mu)}{\mu}}} - \tau \quad (7)$$

222

223 where α are the initial exponential growth rate (aka relative or specific growth rate) of
 224 detritus at detachment or excision, τ is the final exponential decay rate of detritus *sensu*
 225 *stricto* once detrital photosynthesis has ceased, and μ is the time of transition between the two
 226 states when the influence of detrital photosynthesis on decomposition ceases. The influence
 227 of each parameter on the trajectory of k is visualised in Figure 1. Inserting Equation 7 into
 228 Equation 4 yields

229

$$\frac{dm}{dt} = m \left(\frac{\alpha + \tau}{1 + e^{\frac{5(t-\mu)}{\mu}}} - \tau \right) \quad (8)$$

231

232 This equation could be used directly with statistical software that allows solving ordinary
 233 differential equations on the fly. But there is a simpler, generally applicable solution. Rovira
 234 & Rovira (2010) have shown that the logistic function can simply be integrated alongside the
 235 ordinary differential equation to yield a simpler model akin to the integrated form of Equation
 236 3. Since we know our function is bounded by $t = 0$, we use an accumulation function. The
 237 definite integral of Equation 7 is

238

$$\int_0^t k(t) dt = t\alpha + \frac{\mu(\alpha + \tau)}{5} \ln \frac{1 + e^{\frac{5(t-\mu)}{\mu}}}{1 + e^{-5}} \quad (9)$$

240

241 Thus when Equation 8 is integrated as a whole, we get the final form

242

$$m(t) = m_0 e^{t\alpha + \frac{\mu(\alpha + \tau)}{5} \ln \frac{1 + e^{\frac{5(t-\mu)}{\mu}}}{1 + e^{-5}}} \quad (10)$$

244

245 Since decomposition data are best analysed as proportions of initial mass, $m_0 = 1$ which
 246 simplifies the function to one with only three parameters, the now familiar α , μ and τ . The
 247 influence of each parameter on the shape of the curve m describes over time is illustrated in
 248 Figure 1. Note that these parameters may be formally compared within and between studies,
 249 including those that used the traditional model (Equation 3). τ is the exponential decay rate of
 250 detritus *sensu stricto* which is not under the influence of detrital photosynthesis. Since the
 251 traditional model assumes that physiology does not play a role in detrital dynamics, k in

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3 252 Equation 3 also represents the exponential decay rate of detritus *sensu stricto*. Thus τ may be
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5 253 directly compared to k . In the next section I turn this mathematical model into a fully
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7 254 functional statistical model.
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10 256 **The statistical model**

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12 257 Maximum likelihood solutions to nonlinear modelling invariably demand the tedious task of
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14 258 guessing starting values. Rovira & Rovira (2010) followed this approach using the
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16 259 Levenberg-Marquardt algorithm which is more robust to poor starting values than the usual
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18 260 Gauss-Newton algorithm. Despite this advantage, they still needed to manually approximate
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20 261 the starting values before passing them to the algorithm. Bayesian inference using Markov
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22 262 Chain Monte Carlo is more flexible in that there is no practical difference between fitting
23
24 263 linear and nonlinear models (McElreath, 2019). Rather than starting values which can make
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26 264 or break the model, a prior probability distribution is provided, which at worst can lead to
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28 265 inefficient sampling and uncertain prediction. With the increase in computing power, every
29
30 266 laptop can now run Markov Chain Monte Carlo. For these and other reasons Bayesian
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32 267 inference is quickly becoming the *status quo* in applied statistics across disciplines
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34 268 (McElreath, 2019). The most popular statistical language is R (R Core Team, 2022) and the
35
36 269 most popular Bayesian language is Stan (Carpenter et al., 2017) which integrates nicely with
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38 270 R through various packages. Hence I developed the statistical model within this joint
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40 271 framework.
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43 273 Several hierarchical functions are needed to describe a full statistical model. The one to rule
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45 274 them all is the likelihood function. It describes the probability distribution of the data. Most
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47 275 probability distributions are fully described by two parameters, usually the expected value
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49 276 and spread. For the normal distribution this is the mean and variance. Each of these
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51 277 likelihood parameters needs to be a function of the predictor. Finally the functions describing
52
53 278 the likelihood parameters in turn have parameters, which in the Bayesian framework are
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55 279 expressed as probability distribution functions, the priors which once conditioned on the data
56
57 280 turn into posteriors.
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59 281

60 282 The mathematical model (Equation 10) describes the expected value of the likelihood. But
61
62 283 what likelihood? We know that the data are proportions of initial mass that can exceed 1 due
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64 284 to growth caused by detrital photosynthesis. Several distributions are restricted to positive
65
66 285 real numbers, such as the popular gamma distribution. But we can do better by thinking

generatively. The proportions are ratios of mass to initial mass. Mass itself can only take positive real numbers and may be thought of as the sum of cells, growth increments etc. Thus the gamma distribution describes mass well. The ratio of two gamma distributions is a beta prime distribution[§]. Voilà, we have our likelihood!

The beta prime distribution, typically parameterised by the two shape parameters α and β , is most intuitively parameterised in terms of mean $\mu = \frac{\alpha}{\beta - 1}$ and precision $\nu = \beta - 2$, where variance $\sigma^2 = \frac{\mu(1 + \mu)}{\nu}$ (Bourguignon et al., 2021). To reparametrise the likelihood, α and β need to be expressed in terms of μ and ν as $\alpha = \mu(1 + \nu)$ and $\beta = 2 + \nu$. We can either assume ν to be a constant (homoskedasticity) or a function of the predictor (heteroskedasticity). Given that for proportional data $m(0) = 1$ is predetermined, there is no uncertainty at $t = 0$, so ν is theoretically infinite to begin with. But with time comes uncertainty and we expect precision to decline. Sometimes remaining mass is calculated as a proportion of the mean rather than sample-specific initial mass so there is variability at $t = 0$ (e.g. Brouwer, 1996), but even then the variability at $t = 0$ is naturally lowest. Since $\nu > 0$ an appropriate model is

$$\nu(t) = (\epsilon - \theta)e^{-\lambda t} + \theta \quad (11)$$

where ϵ is ν at $t = 0$, λ is the rate of exponential decay in ν , and θ is ν as $t \rightarrow \infty$, inspired by its representation of Thánatos. Combining the beta prime likelihood and Equations 10 and 11 yields the full statistical model:

$$m \sim \text{Beta}'(\mu_m(1 + \nu), 2 + \nu) \quad (12)$$

$$\mu_m = e^{t\alpha + \frac{\mu(\alpha + \tau)}{5} \ln \frac{1 + e^{\frac{5(t - \mu)}{\mu}}}{1 + e^{-5}}}$$

$$\nu = (\epsilon - \theta)e^{-\lambda t} + \theta$$

[§] Technically this is only true if both gamma distributions share the same rate or scale, i.e. size of summand, and are independent. We can assume the former because both measures of mass can be broken down into the same components, e.g. cells. It is difficult to assume the latter because the fundamental assumption of the model is that the change in mass depends on mass (Equation 1, 3). Nonetheless, beta prime remains the best approximation of the true data distribution I can think of.

1
2
3 311 $\alpha \sim N(\dots), \text{Exp}(\dots)$ or $\text{Gamma}(\dots)$
4
5 312 $\mu, \tau, \epsilon, \lambda, \theta \sim \text{Exp}(\dots)$ or $\text{Gamma}(\dots)$
6
7 313
8
9 314 where m is the proportion of remaining mass and μ_m is the mean of m , the remaining
10 315 parameters being the same as in Equations 10 and 11. α may have a normal prior since it can
11 316 take positive (initial growth) or negative (initial decomposition) values. However, in some
12 317 cases it may be desirable to enforce initial growth with an exponential or gamma prior. All
13 318 other parameters are restricted to positive values, so the exponential or gamma distribution is
14 319 most suitable. I found generally suitable priors to be $\epsilon \sim \text{Gamma}(4, 10^{-4})$, $\lambda \sim \text{Exp}(1)$ and θ
15 320 $\sim \text{Gamma}(4, 0.008)$, especially because these parameters are a bit abstract. I use these
16 321 priors in all examples in the next section. For α , μ and τ I cannot recommend such generally
17 322 applicable priors. As is usual with Bayesian workflow (McElreath, 2019), priors have to be
18 323 chosen based on prior simulation. Furthermore, α , μ and τ are biologically meaningful and
19 324 can thus be strongly informed by the literature. Some of the simplest examples that may be a
20 325 reasonable starting point are $\alpha \sim N(0, 0.01)$, $\mu \sim \text{Exp}(0.1)$ and $\tau \sim \text{Exp}(10)$. But in most
21 326 cases these will have to be further constrained to ensure smooth sampling. The best next step
22 327 would be to use a tighter normal prior for α , a gamma prior centred on half the range of t for
23 328 μ and something along the lines of $\tau \sim \text{Gamma}(4, 40)$ which has mean 0.1 and standard
24 329 deviation 0.05. But in most cases you will be able to do far better.
25 330
26 331 Other likelihoods may also yield reasonable predictions if Equation 12 cannot be
27 332 implemented in your statistical framework of choice. The gamma distribution may only get
28 333 extreme values wrong due to its differently shaped tail and the normal distribution may still
29 334 provide valid predictions of the mean even though it is not constrained to positive real
30 335 numbers. If necessary, the model can also be simplified by assuming homoskedasticity,
31 336 which may have minimal impact on mean predictions albeit inflating uncertainty near $t = 0$
32 337 and potentially underestimating uncertainty as $t \rightarrow \infty$.
33 338
34 339 There are some final considerations for the statistical model that are purely computational.
35 340 Stan uses Hamiltonian Monte Carlo to explore probability space. Some of the expressions I
36 341 have used in Equations 10 and 11 are easier to read but not friendly to the sampler. These can
37 342 be improved for computational efficiency using the quotient and product rules of logarithms
38 343 as

344

$$m(t) = e^{t\alpha + \frac{\mu(\alpha + \tau)}{5}(\ln(1 + e^{\frac{5(t - \mu)}{\mu}}) - \ln(1 + e^{-5}))} \quad (13)$$

346

$$v(t) = e^{\ln(\epsilon - \theta) - \lambda t + \theta} \quad (14)$$

348

349 The $\ln(1 + e^x)$ expressions in Equation 13 are further simplified in Stan using the wrapper

350 function `loglp_exp` which is numerically stabler. An equivalent function doesn't exist in R

351 but can be written as `loglp_exp <- function(x) ifelse( x > 0 , x +`

352 `loglp( exp(-x) ) , loglp( exp(x) ) )`.

353

354 In terms of software, I use R v4.5.1 with the tidyverse package family v2.0.0 (Wickham et

355 al., 2019) within the integrated development environment RStudio v2025.09.0+387 (RStudio

356 Team, 2022) and run Stan with the R interface cmdstanr v0.9.0 (Gabry et al., 2024) via

357 CmdStan v2.37.0 (Lee et al., 2017). All models have 8 Markov chains with 10^4 warmup and

358 sampling iterations each. To assess quality, I use $\hat{R}$ scores and visually scrutinise trace rank

359 and pair plots with bayesplot v1.14.0 (Gabry et al., 2019). All reported results are posterior

360 probabilities and derived central tendencies and intervals calculated with tidybayes v3.0.7

361 (Kay, 2024). See github.com/lukaseamus/limbodeco for further details.

362

363 In the next section I will apply Equation 12 to four ecological effects in six published studies

364 (Bourguès et al., 1996, Brouwer, 1996, Hamersley et al., 2015, de Bettignies et al., 2020,

365 Frontier et al., 2021, Frontier et al., 2022). I either received data from the lead author

366 (Frontier et al., 2021, Frontier et al., 2022) or digitised figures (Bourguès et al., 1996,

367 Brouwer, 1996, Hamersley et al., 2015, de Bettignies et al., 2020) using WebPlotDigitizer

368 v4.6 (Rohatgi, 2022). When several samples were collected at each timepoint, figures only

369 yielded mean and standard deviation of remaining mass. The typical approach in such cases

370 is measurement error modelling to account for the uncertainty of each mean so that means

371 with larger standard deviations carry less weight in influencing the prediction (McElreath,

372 2019). However, at each timepoint de Bettignies et al. (2020) collected 5 samples,

373 summarised as mean and standard deviation, for one treatment and only 1 sample for the

374 other, resulting in a mixture of sample sizes and data levels. A measurement error model is

375 not applicable in such a case. Furthermore, I attempted building beta prime, gamma and

376 lognormal measurement error models and they all result in poor sampling when paired with

the beta prime likelihood. Only a zero-truncated normal measurement error distribution was feasible but this does not necessarily represent the data distribution. I therefore simulated observations from summary statistics as random draws from the beta prime distribution in R using the *rbetapr* function from *extraDistr* v1.10.0 (Wolodzko, 2023) as

```
rbetapr (
  n ,
  mean * ( 1 + mean * ( 1 + mean ) / sd^2 ) ,
  2 + mean * ( 1 + mean ) / sd^2
)
```

where *n*, *mean* and *sd* are the reported sample size, sample mean and sample standard deviation. I derived this alternative parameterisation of the beta prime distribution by inserting $v = \frac{\mu(1 + \mu)}{\sigma^2}$ into $\alpha = \mu(1 + v)$ and $\beta = 2 + v$ (Bourguignon et al., 2021).

Examples

Dead or alive

Brouwer (1996) provides a nice example of detrital photosynthesis influencing decomposition of the feathery brown seaweed *Desmarestia anceps* at Signy Island in Antarctica. She killed some of the detritus by heating to 50°C for 10–15 min prior to *in situ* decomposition (Brouwer, 1996). The resulting effect is remarkable but Brouwer (1996) was unable to quantify it due to lack of a suitable model at the time, so only provided an analysis of variance *F* statistic for change over time in both treatments and described the divergence in the decomposition trajectories.

The statistical model outlined in Equation 12 given this dataset and priors $\alpha \sim N(- 0.005 , 0.003)$, $\mu \sim \text{Gamma}(\frac{200^2}{150^2} , \frac{200}{150^2})$ and $\tau \sim \text{Gamma}(\frac{0.1^2}{0.05^2} , \frac{0.1}{0.05^2})$ is visualised in Figure 2. Predictions across *t* for new observations of *m* based on the full likelihood ($\tilde{m}$) are in the top panel, followed by the likelihood parameters *v* and μ_m , and *k* in the bottom panel. One could also overlay the probability distributions for the basal parameters α , μ and τ and ϵ and θ onto the plots of *k* and *v* respectively to present the full hierarchy of the model. To avoid clutter I have opted to only display μ , since this is the key parameter, marking the half-life of detrital photosynthesis, which, unlike α , τ , ϵ and θ , cannot be guessed by looking at

predictions of k and v . Summaries of posterior probability distributions for the basal parameters describing μ_m and k are given in Table 1. I decided to pool τ , ϵ and θ across treatments. The experimental duration is not long enough to inform τ for the control treatment and it is reasonable to assume that untreated detritus decomposes at the same rate as pre-killed detritus once physiology has ceased. There is also no reason to suspect ϵ and θ to vary across treatments, since ϵ is only governed by the variability in initial experimental mass, which should be unaffected by treatment, and θ goes hand in hand with τ .

The model suggests that pre-killed *D. anceps* detritus decomposes much faster than the control (Figure 2). Untreated detritus has an initial relative growth rate of 0.037 ± 0.017 % d^{-1} (mean $\pm$ standard deviation, $P_{\alpha > 0} = 0.99$) while pre-killed detritus has an initial decay rate of 0.6 ± 0.29 % d^{-1} ($P_{\alpha < 0} = 0.98$) (Table 1). Detrital photosynthesis of *D. anceps* only ceases to influence decomposition after 778 ± 108 d (Table 1). Pre-killed detritus starts decomposing at the ultimate rate 750 ± 106 d earlier ($P_{\Delta\mu > 0} = 1$) (Table 1). Without physiology, detritus of *D. anceps* therefore decomposes in about 3.6 ± 0.56 % of the expected time.

Senescence

Hamersley et al. (2015) and de Bettignies et al. (2020) both studied the effect of tissue senescence on kelp decomposition. Hamersley et al. (2015) collected attached fresh and senescent fronds of the giant kelp *Macrocystis pyrifera* at Catalina Island in California along with naturally detached wrack and raft material which they regarded as detritus *sensu stricto*. de Bettignies et al. (2020) collected tissue from above and below the May cast abscission border of the kelp *Laminaria hyperborea* at Roscoff in Brittany. Their senescent blade thus represents tissue that the plant is in the process of abscising. Hamersley et al. (2015) refrained from fitting a model but de Bettignies et al. (2020) are among the few that fit Equation 3 to kelp decomposition data.

My models for *M. pyrifera* (Figure 3a) and *L. hyperborea* (Figure 3b) were both given priors $\alpha \sim N(0, 0.01)$ and $\mu \sim \text{Exp}(0.1)$. But since de Bettignies et al. (2020) reported k values (Equation 3) of 0.0366 d^{-1} for senescent and 0.0107 d^{-1} for fresh lamina, I was able to better constrain the τ prior for this dataset. I therefore used $\tau \sim \text{Gamma}(\frac{0.2^2}{0.1^2}, \frac{0.2}{0.1^2})$ for *M. pyrifera* and $\tau \sim \text{Gamma}(\frac{0.0366^2}{0.02^2}, \frac{0.0366}{0.02^2})$ for *L. hyperborea*. For clarity I here only show predictions for k and μ_m with posterior probability distributions for μ , but the model provides the full set

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3 443 of information as in Figure 2. While species is constant across treatments in each study, the
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5 444 detritus has various natural origins rather than having been subjected to different
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7 445 experimental treatments as in the previous example. So I allowed all parameters except for ϵ
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9 446 to vary by treatment. I still pooled ϵ because, as is usual but unlike the previous example,
10 447 there are no data to inform variability at $t = 0$ in these datasets.
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13 449 Both models state that senescent fronds decompose faster than fresh ones, and detached ones
14 450 decompose faster still (Figure 3). Initial decomposition rates for all treatments of *M. pyrifera*
15 451 (Table 1) are not distinguishable from zero ($P_{\alpha < 0} = 0.53\text{--}0.74$) or from one another ($P_{\Delta\alpha > 0} =$
16 452 $0.53\text{--}0.58$). However, the drop in k takes place 1.3 ± 0.81 d ($P_{\Delta\mu > 0} = 0.97$) earlier for
17 453 detached than senescent tissue for which in turn it happens 3.6 ± 1.3 d ($P_{\Delta\mu > 0} = 1$) earlier
18 454 than for fresh tissue (Figure 3a). The final decomposition rate for fresh tissue is 2.8 ± 1.2 %
19 455 d^{-1} ($P_{\Delta\tau > 0} = 0.99$) and 0.47 ± 1.2 % d^{-1} ($P_{\Delta\tau > 0} = 0.65$) greater than for senescent and
20 456 detached tissue. Fresh *L. hyperborea* has an initial growth rate of 1 ± 0.6 % d^{-1} ($P_{\alpha > 0} = 0.97$)
21 457 while senescent tissue of the same plant has an initial decay rate of 0.27 ± 1 % d^{-1} ($P_{\alpha < 0} =$
22 458 0.6) (Table 1). Detrital photosynthesis in fresh tissue apparently ceases relatively early—after
23 459 only 10 ± 3.1 d which is 6.7 ± 4.9 d ($P_{\Delta\mu > 0} = 0.93$) after the senescent tissue—but its final
24 460 decomposition rate is less than half that of senescent tissue ($\Delta\tau = 2 \pm 0.38$ % d^{-1} , $P_{\Delta\tau > 0} = 1$)
25 461 (Table 1). Due to this apparently relatively minor influence of detrital photosynthesis,
26 462 decomposition rates are ultimately similar to those reported by (de Bettignies et al., 2020):
27 463 0.013 ± 0.0011 d^{-1} vs. 0.0107 d^{-1} for fresh and 0.032 ± 0.0036 vs. 0.0366 d^{-1} for senescent *L.*
28 464 *hyperborea* (Table 1).
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43 466 **Light**

44 467 The effect of depth on decomposition rates has repeatedly been evidenced for *Laminaria*
45 468 species (Frontier et al., 2021, Frontier et al., 2022). Detritus of *L. hyperborea* and *L.*
46 469 *ochroleuca* was either exposed to three simulated depths *ex situ* at Roscoff in Brittany
47 470 (Frontier et al., 2021) or deployed at three depths *in situ* across the Channel at Plymouth in
48 471 Devon (Frontier et al., 2022). Photophysiological measurements across detrital age have
49 472 clarified that the amount of light available to detrital photosynthesis is the mechanism behind
50 473 this effect (Frontier et al., 2021, Frontier et al., 2022). However, detrital photosynthesis and
51 474 decomposition have not been quantitatively linked. Furthermore remaining mass was
52 475 analysed using linear models which don't do a very good job given the nature of the data
53 476 (Frontier et al., 2021, Frontier et al., 2022).

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478 Given identical priors $\alpha \sim \text{Exp}(100)$, $\mu \sim \text{Gamma}(\frac{50^2}{30^2}, \frac{50}{30^2})$ and $\tau \sim \text{Gamma}(\frac{0.1^2}{0.05^2}, \frac{0.1}{0.05^2})$,
 479 the results for both datasets are shown in Figure 4. Note that I am here making use of the
 480 possibility of constraining α to positive real numbers. All experimental tissue was excised
 481 from fresh parts of the lamina (Frontier et al., 2021, Frontier et al., 2022) so we expect initial
 482 growth. Due to the complexity of this dataset with three predictor variables, I only show $\tilde{m}$,
 483 the posterior we would use to ultimately make predictions. In both cases I let α vary by
 484 species but not treatment, since there is no reasonable way for treatment to affect initial
 485 growth, and μ vary by species and treatment. As with α , for the first dataset I let τ vary by
 486 species but not by treatment following the rationale outlined above that detritus of the same
 487 origin can be expected to ultimately decompose at the same rate. For the second dataset I
 488 pooled τ across species and treatment since the experimental duration is not long enough to
 489 inform τ for *L. hyperborea* (Figure 4b). A more elegant solution would be partial pooling
 490 using multilevel modelling but that is beyond the scope of this paper. ϵ again was completely
 491 pooled and θ followed the pooling of τ for reasons outlined before.

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493 Both models suggest that decomposition is faster at greater depths (Figure 4) for reasons
 494 outlined above. Decomposition also tends to be faster for *L. ochroleuca* than *L. hyperborea*
 495 (Figure 4), which is well established (Wright et al., 2022), but this does not hold true at
 496 greater depths (Figure 4a). Initial relative growth rates (Table 1) are not different between
 497 species for the first model ($P_{\Delta\alpha > 0} = 0.51$) but are $0.34 \pm 0.13 \% \text{ d}^{-1}$ ($P_{\Delta\alpha > 0} = 1$) greater for *L.*
 498 *ochroleuca* than *L. hyperborea* in the second model. In the first model, detrital photosynthesis
 499 of *L. hyperborea* ceases $175 \pm 38 \text{ d}$ ($P_{\Delta\mu > 0} = 1$) later at the surface but $50 \pm 17 \text{ d}$ ($P_{\Delta\mu > 0} = 1$)
 500 and $12 \pm 3.9 \text{ d}$ ($P_{\Delta\mu > 0} = 1$) earlier at 15 and 30 m. Such a reversal in physiological persistence
 501 with depth is not evident in the second model, where *L. hyperborea* detritus photosynthesises
 502 $89 \pm 35 \text{ d}$ ($P_{\Delta\mu > 0} = 1$), $81 \pm 32 \text{ d}$ ($P_{\Delta\mu > 0} = 1$) and $27 \pm 17 \text{ d}$ ($P_{\Delta\mu > 0} = 1$) longer at 0.5, 1.5 and
 503 3 m, probably due to the lesser depth range. In both models the maximal negative effect of
 504 depth on the longevity of detrital photosynthesis is stronger for *L. hyperborea* ($\Delta\mu = 87 \pm 37$
 505 $- 230 \pm 38 \text{ d}$, $P_{\Delta\mu > 0} = 0.99-1$) than *L. ochroleuca* ($\Delta\mu = 25 \pm 12 - 43 \pm 6.4 \text{ d}$, $P_{\Delta\mu > 0} = 1$). So
 506 the detrital physiology of *L. hyperborea* is less tolerant to low light, which agrees with
 507 observed kelp forest zonation (Wright et al., 2022). The first model, with which it was
 508 possible to quantify differences in τ (Table 1), suggests that *L. ochroleuca* ultimately
 509 decomposes $0.6 \pm 0.89 \% \text{ d}^{-1}$ ($P_{\Delta\tau > 0} = 0.75$) faster than *L. hyperborea*, independent of

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3 510 physiology. However, contrasting detrital physiology is clearly the main driver of the
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5 511 interspecific difference in decomposition timescales, as has been previously hinted at (Wright
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7 512 et al., 2022).

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10 514 **Season**

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12 515 Bourguès et al. (1996) provide perhaps the only data of macroalgal decomposition across all
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14 516 seasons. They studied decomposition of the ephemeral green seaweed *Ulvaria obscura* in the
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16 517 intertidal Arcachon Bay in France. Bourguès et al. (1996) collected fresh *U. obscura* but
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18 518 buried it below 1 cm of sediment where anoxia apparently persisted and there clearly was no
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20 519 light for detrital photosynthesis. Given these experimental conditions, it would be surprising
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22 520 to see decomposition affected by physiology. Yet Bourguès et al. (1996) were faced with the
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24 521 challenge of having to model decomposition trajectories of sigmoidal appearance for which
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26 522 they chose the logistic function. Excised macroalgae can survive exceptional periods of
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28 523 darkness (Wright et al., 2024) so it may be that saprotrophs were initially inhibited by
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30 524 chemical defence despite the absence of photosynthesis. Even if this was not the case because
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32 525 anoxia immediately killed the seaweed, and the model therefore is conceptually wrong, it is
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34 526 still useful because it avoids having to fit fundamentally incomparable models as Bourguès et
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36 527 al. (1996) have done.

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38 529 Given this dataset and priors $\alpha \sim \text{Exp}(100)$, $\mu \sim \text{Gamma}(\frac{25^2}{10^2}, \frac{25}{10^2})$ and $\tau \sim \text{Gamma}(\frac{0.1^2}{0.05^2}, \frac{0.1}{0.05^2})$, $\tilde{m}$ is visualised in Figure 5. Again I make use of constraining α to positive values
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40 530 because the seaweed was initially alive. Season, which for all intents and purposes is
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42 531 temperature, affects detrital physiology as well as other aspects that may drive decomposition
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44 532 post-physiology such as saprotroph metabolic rates (Filbee-Dexter et al., 2022, Wright et al.,
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46 533 2024). So all parameters are expected to be affected by season. I therefore only pooled ϵ
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48 534 across treatments for reasons outlined above.

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51 537 According to the model decomposition is generally faster in warmer seasons (Figure 5). This
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53 538 agrees well with the known effect of temperature on decomposition (Filbee-Dexter et al.,
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55 539 2022, Wright et al., 2024). Initial growth rate is not strikingly different between any two
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57 540 seasons ($P_{\Delta\alpha > 0} = 0.54\text{--}79$), perhaps unsurprisingly given data sparsity immediately after the
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59 541 start of the experiment (Figure 5). Differences are more pronounced in the decline of k over
60 542 time. k shifts at a similar time in autumn and winter ($P_{\Delta\mu > 0} = 0.52$) but does so 9 ± 3.6 d ($P_{\Delta\mu}$

$> 0 = 1$) and 19 ± 3.3 d ($P_{\Delta\mu > 0} = 1$) earlier in spring and summer than in autumn (Table 1). This may be the detrimental effect of increased temperature on detrital physiology (Wright et al., 2024) since spring and summer are generally warmer than autumn and winter, unless we believe the experimental conditions immediately killed the seaweed. The final value of k is 6 ± 6.4 % d⁻¹ ($P_{\Delta\tau > 0} = 0.84$), 2.8 ± 5.5 % d⁻¹ ($P_{\Delta\tau > 0} = 0.73$) and 5 ± 6 % d⁻¹ ($P_{\Delta\tau > 0} = 0.81$) lower—i.e. decomposition is faster and τ is greater—in spring, summer and autumn than in winter (Table 1). This is the effect of seasonal temperature differences on factors other than detrital physiology. Again the physiological effect, albeit putative in this case, is more pronounced, as has been suggested in the context of temperature (Wright et al., 2024).

Conclusion

Macroalgal decomposition has neither been consistently nor reliably modelled. But a realistic model is the key to understanding macroalgal blue carbon. I have presented the first decomposition model tailored to macroalgae by taking into account the theory of detrital photosynthesis. This is made possible by treating the exponential decay rate as a function of time rather than a constant, a function which is informed by the latest understanding of detrital photosynthesis. My model only has three parameters which describe the initial relative growth rate (α), the half-life of detrital photosynthesis (μ), and the final exponential decay rate (τ) which is directly comparable to that of the classic model (k). I have shown that these parameters are successfully estimated and meaningful for empirical data from a variety of contexts. Given a set of parameter estimates, the model can reliably predict new observations, even beyond the duration of the experiment. Although I have written the model in R with Stan, it can be implemented in any statistical framework that allows robust nonlinear modelling. Failing to account for detrital photosynthesis almost invariably leads to underestimation of macroalgal decomposition. I therefore expect that this model will increase estimates of macroalgal decomposition and thus have major implications for macroalgal blue carbon accounting. I will be using this model to clarify the role of kelp forests in CO₂ removal and hope others find it useful too.

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47 **Tables**

48 736 **Table 1.** Macroalgal decomposition parameter estimates (mean ± standard deviation) for
49 737 Equation 10 (μ_m in Equation 12) based on 8×10^4 posterior samples. Units are given in
50 738 parentheses. For readability the exponential rates α and τ are given as daily percentages, but
51 739 they are modelled as daily proportions. Pooled estimates within a study are marked with the
52 740 same symbol.

Species	Treatment	α (% d ⁻¹)	μ (d)	τ (% d ⁻¹)
Brouwer 1996				
<i>Desmarestia anceps</i>	Control	0.037 ± 0.017	778 ± 108	7.2 ± 0.91*
	Pre-killed	-0.6 ± 0.29	28 ± 4.1	7.2 ± 0.91*

Hamersley <i>et al.</i> 2015				
<i>Macrocystis pyrifera</i>		Fresh	-0.31 ± 0.55	5.9 ± 1
		Senescent	-0.19 ± 1	2.3 ± 0.75
		Detached	-0.076 ± 1	0.94 ± 0.3
de Bettignies <i>et al.</i> 2020				
<i>Laminaria hyperborea</i>		Fresh	1 ± 0.6	10 ± 3.1
		Senescent	-0.27 ± 1	3.3 ± 3.7
Frontier <i>et al.</i> 2021				
<i>Laminaria hyperborea</i>		0 m	$0.03 \pm 0.017^*$	259 ± 48
		15 m	$0.03 \pm 0.017^*$	27 ± 9.2
		30 m	$0.03 \pm 0.017^*$	1.9 ± 1.2
<i>Laminaria ochroleuca</i>		0 m	$0.037 \pm 0.032^\dagger$	57 ± 6.5
		15 m	$0.037 \pm 0.032^\dagger$	80 ± 18
		30 m	$0.037 \pm 0.032^\dagger$	14 ± 3.7
Frontier <i>et al.</i> 2022				
<i>Laminaria hyperborea</i>		0.5 m	$0.24 \pm 0.042^*$	105 ± 22
		1.5 m	$0.24 \pm 0.042^*$	94 ± 20
		3 m	$0.24 \pm 0.042^*$	42 ± 12
<i>Laminaria ochroleuca</i>		0.5 m	$0.58 \pm 0.13^\dagger$	43 ± 9.2
		1.5 m	$0.58 \pm 0.13^\dagger$	35 ± 8.7
		3 m	$0.58 \pm 0.13^\dagger$	21 ± 3.8
Bourguès <i>et al.</i> 1996				
<i>Ulvaria obscura</i>		Spring	0.43 ± 0.4	14 ± 2.2
		Summer	0.7 ± 0.71	4.4 ± 1.8
		Autumn	0.6 ± 0.33	23 ± 2.8
		Winter	0.29 ± 0.3	24 ± 7.3

Figure captions

Figure 1. The mathematical model. Some reasonable parameter values for α , μ and τ and how they independently determine the behaviour of the exponential decay rate k (Equation 7) and the proportion of remaining mass m (Equation 10) over time t .

Figure 2. The statistical model. Example 1: the effect of physiology on the decomposition of *Desmarestia anceps*, shown as the proportion of remaining mass over time t (data from Brouwer, 1996). Data are plotted alongside predictions of new observations ($\tilde{m}$), the precision (v), the mean (μ_m) and the exponential decay rate (k) (Equation 12). Points with error bars denote mean $\pm$ standard deviation of 5 samples. Lines and intervals are medians and the central 50, 80 and 90% of posterior probability.

Figure 3. The statistical model. Example 2: the effect of senescence on the decomposition of *Macrocystis pyrifera* (**a**, data from Hamersley *et al.*, 2015) and *Laminaria hyperborea* (**b**, data from de Bettignies *et al.*, 2020), shown as the proportion of remaining mass over time t . Data are plotted alongside predictions of the mean (μ_m) and the exponential decay rate (k) (Equation 12). Points with error bars denote mean $\pm$ standard deviation of 5 samples. Points

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without error bars are 1 sample. Lines and intervals are medians and the central 50, 80 and 90% of posterior probability.

Figure 4. The statistical model. Example 3: the effect of light attenuation with depth on the decomposition of *Laminaria hyperborea* and *Laminaria ochroleuca ex situ* (**a**, data from Frontier et al., 2021) and *in situ* (**b**, data from Frontier et al., 2022), shown as the proportion of remaining mass over time t . Data are plotted alongside predictions of new observations ($\tilde{m}$, Equation 12). Lines and intervals are medians and the central 50, 80 and 90% of posterior probability.

Figure 5. The statistical model. Example 4: the effect of season on the decomposition of *Ulvaria obscura* (data from Bourguès et al., 1996). Data are plotted alongside predictions of new observations ($\tilde{m}$, Equation 12). Lines and intervals are medians and the central 50, 80 and 90% of posterior probability.

Acknowledgements

Thanks go to Andy Foggo and Nadia Frontier for inspiring discussions over the years and Nadia for sharing her data with me. Thanks also to Pere Rovira for the email exchange and Bojana Manojlovic for the chat.

Funding

This work was supported by the Forrest Research Foundation Scholarship and Australian Government Research Training Program Scholarship. Open access publishing was facilitated by The University of Western Australia, as part of the Wiley–The University of Western Australia agreement via the Council of Australian University Librarians.

Conflicts of interest

I declare no conflicts of interest.

Data and availability

Data and code to reproduce this paper and apply my model to your own data are available at github.com/lukaseamus/limbodeco. Please cite this paper as well as the model (Wright, 2025).









